

Project Design Phase-II

Data Flow Diagram & User Stories

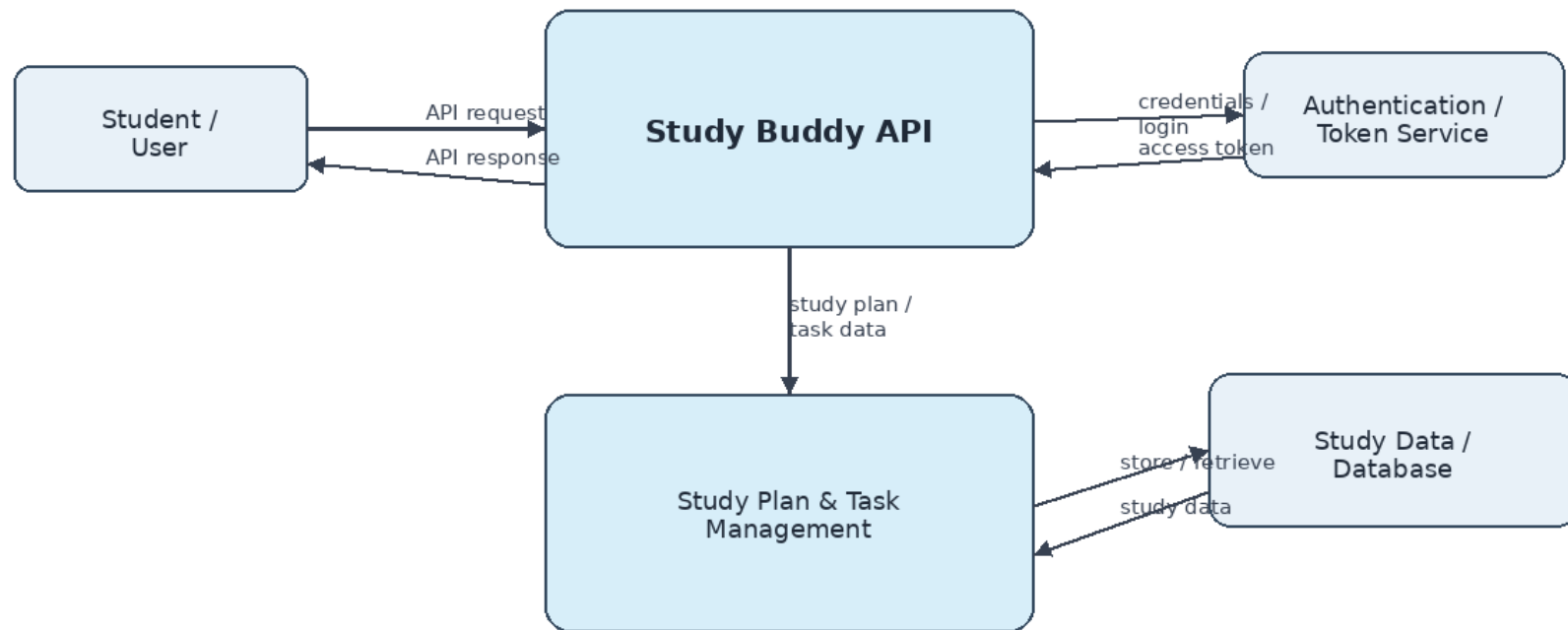
Date	30 September 2026
Team ID	SWTID-2026-5226
Project Name	Study Buddy API
Maximum Marks	4 Marks

Data Flow Diagrams:

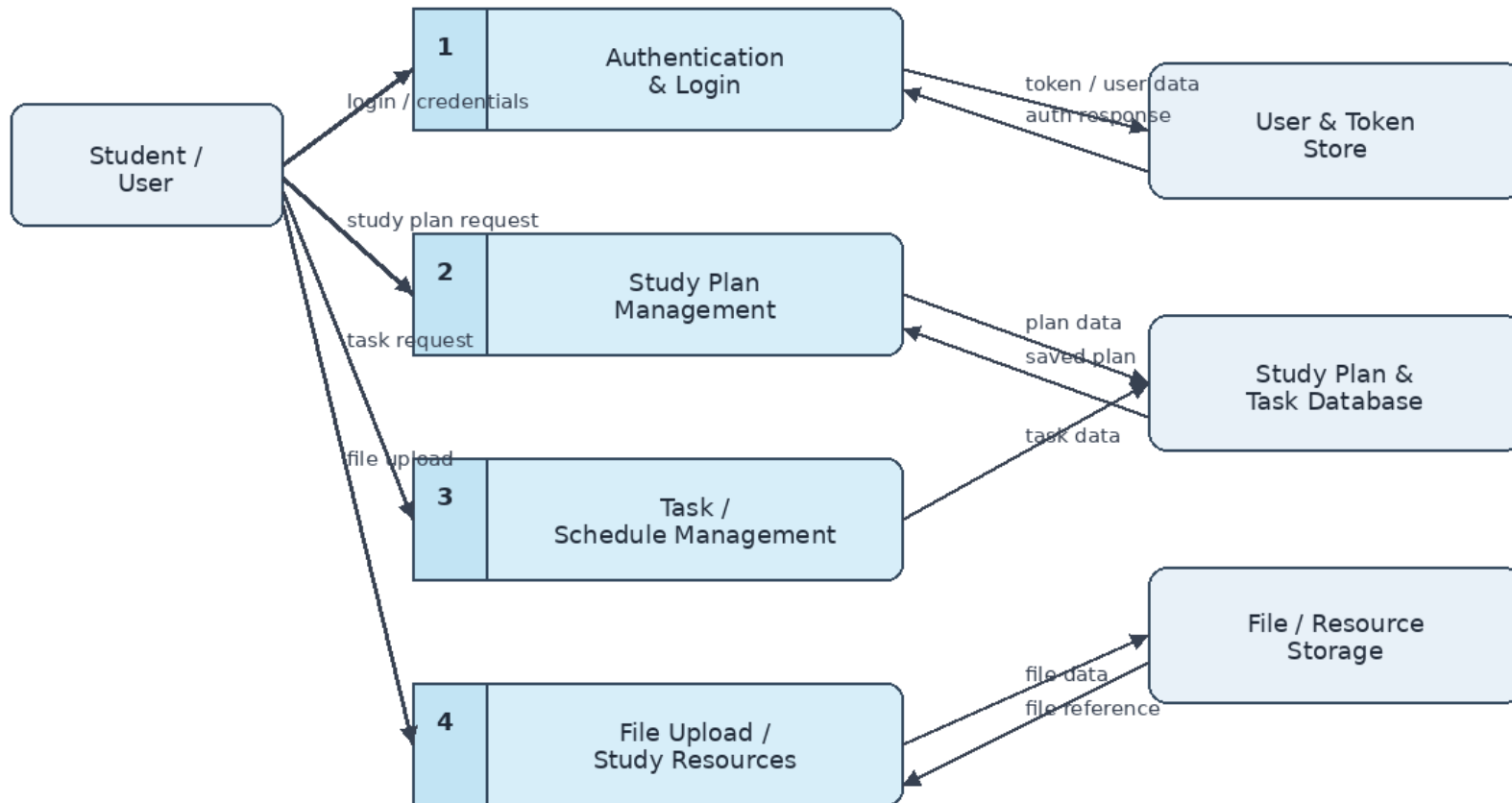
A Data Flow Diagram (DFD) represents how information moves through the Study Buddy API. It shows user requests, authentication, study-plan and task operations, file uploads, and storage used by the application.

Study Buddy API: Data Flow

Study Buddy API - DFD Level 0

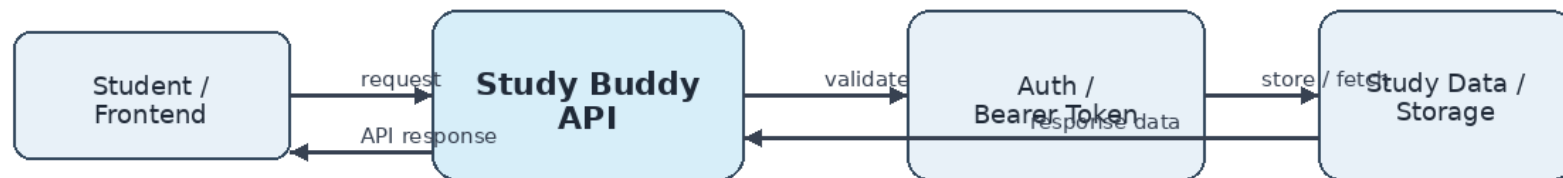


Study Buddy API - DFD Level 1



Study Buddy API - Data Flow

- 1 User sends an API request with login details or study data.
- 2 API validates the request and authenticates the user.
- 3 Access token is issued and used as a Bearer Token for protected APIs.
- 4 Study Plan, Task, and File Upload APIs process the requested operation.
- 5 Study data is stored/retrieved and the API returns a response to the user.



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Student / User	Registration	USN-1	As a user, I can register for Study Buddy using my email and password.	Account is created successfully and the user can access the application.	High	Sprint-1
	Login	USN-2	As a user, I can log into Study Buddy using my registered email and password.	Valid credentials return a successful login response.	High	Sprint-1
	Authentication	USN-3	As a user, I can receive an access token after successful login.	A valid Bearer Token is returned and can be used for protected API requests.	High	Sprint-1
	Study Plan	USN-4	As a user, I can create a study plan through the Study Plan API.	The study plan is validated and saved successfully.	High	Sprint-1
	Study Plan	USN-5	As a user, I can view and update my study plan.	The latest study-plan details are returned and updates are saved.	High	Sprint-2
	Tasks	USN-6	As a user, I can add, update, and delete study tasks.	The requested task operation returns a successful API response.	High	Sprint-2
	File Upload	USN-7	As a user, I can upload a study file using the File Upload API.	The file is uploaded successfully and a valid file reference is returned.	Medium	Sprint-2
	Study Resources	USN-8	As a user, I can access my uploaded study resources.	Only the authenticated user's available resources are returned.	Medium	Sprint-2
	Progress Tracking	USN-9	As a user, I can view my study progress and completed tasks.	The API returns current progress information for the user.	Medium	Sprint-3
	Profile	USN-10	As a user, I can view and update my Study Buddy profile.	Profile details are updated and returned correctly.	Low	Sprint-3